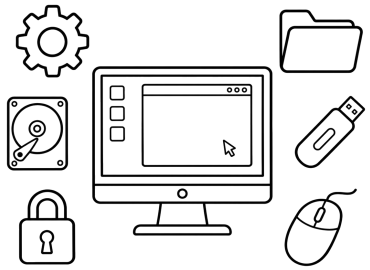
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 10 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 5
Operating Systems



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 5 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - what to check in Practicals 1 - 4, safety, a quiz, a practical test
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Unit 5 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links, homework ideas and teaching tips

★ BEFORE THE FIRST LESSON

- ☐ Command Prompt and Task Manager open on every lab computer.
- ☐ File name extensions can be shown in File Explorer (View tab).
- ☐ Two or three pen drives with no important data are ready, scanned for viruses.

"Students remember what they do on the computer, not only what they read about it."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Complete the software tree	8
	B Fill in the blanks	8
	C Put the booting steps in order	5
	D Answer these	4
	E Which part of the computer system?	8
Chapter 2	A Which type of operating system?	8
	B Label the screens	8
	C Match them	6
	D Choose the correct answer	4
	E True or false	6
Chapter 3	A Which management task?	8
	B Read the Properties window	6
	C Sequential or random access?	6
	D Complete the file systems table	5
	E Read the file names	10
Chapter 4	A Read the drives window	8
	B Where is the file?	4
	C Which utility?	8
	D Name the disks	5
	E True or false	8
	F Crossword	10
End	Unit 5 Test	40
TOTAL		**183**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Start the unit by switching on a computer in front of the class. Ask: what do you see first?
- Write the booting steps on the board as a chain: POWER ON, POST, BOOTSTRAP LOADER, OS LOADED.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (33)	Ch 2 (32)	Ch 3 (35)	Ch 4 (43)	Test (40)	Total (183)
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
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24							
25							
26							
27							
28							
29							

TEACHER GUIDE

Practicals at a Glance

Textbook pages 138 - 157 and the Activity on 156 - 157

Practical	Students do	Check that ...
1 (pp. 138 - 141)	watch the computer boot; name three application software and the OS	they say when the firmware text shows and when the OS takes control; Windows is proprietary.
2 (p. 156)	Command Prompt: date, time, dir, cls, exit; the WIMP parts of the desktop	the output looks like Figure 5.3; they point to a window, an icon, a menu and the pointer.
3 (pp. 156 - 157)	Task Manager; plug and play; Windows key + L; file name extensions, Properties and a backup copy	Notepad shows as a new process; the driver message appears; the Properties answers are right; drive C: is NTFS.
4 (p. 157)	the drives in This PC, a pen drive, drive Properties, Analyze, five utilities	the letters are written down; only Analyze is clicked; five utilities with their uses.

! SAFETY AND MATERIALS

- Practical 1: start the computers together so that everyone sees the booting steps. Do not open the BIOS set-up.
- Practicals 3 and 4: students only look in Task Manager and the disk tools. They never end a process, format, partition or defragment a drive.
- Two or three pen drives with no important data, scanned with the antivirus software first.

Textbook links

Workbook	Textbook pages	Activity questions
Chapter 1 - The computer system and software	138 - 141, 155	-
Chapter 2 - Types, services and user interfaces	141 - 144	4 - 9
Chapter 3 - Managing the hardware and files	144 - 149	1 - 3, 10, 12, 14
Chapter 4 - Utility software, drives and disks	150 - 155	11, 13, 15
Unit 5 in One Page	138 - 155	-

★ TEACHING TIPS

- Remind students that firmware is not on the hard disk: the booting instructions live in the ROM.
- Use the school for liveware: the students, the teachers and the computer lab assistant.
- Draw Figure 5.2 as three rings on the board and let students place the software names in them.
- Ask which OS the phones at home use. Sort the answers into proprietary and free and open source.
- Explain multi-tasking with music playing while a student types a letter on the same computer.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. Name the four parts of a computer system.	**hardware, firmware, software, liveware**
2. Where is the BIOS stored?	**in ROM**
3. What does POST stand for?	**Power-On Self-Test**
4. Name the three language translators.	**assembler, compiler, interpreter**
5. Name a free and open source OS.	**Ubuntu, Fedora, Android, Linux**
6. Which type of OS does an ATM use?	**a real time OS**
7. What does WIMP stand for?	**Windows, Icons, Menus, Pointer**
8. What is a process?	**a running program**
9. What type of file is marks.xlsx?	**an Excel file**
10. What is done to a new hard disk first?	**partition, then format**

> PRACTICAL TEST - 30 MINUTES, 20 MARKS (OPTIONAL)

Use the computers of Practicals 2 and 3. Students work alone; give the marks while you watch.

Step	The student must ...	Marks
1	Open Command Prompt. Type **date** and **time** and show the output.	3
2	Open Task Manager. Name two processes that are running.	3
3	Show the file name extensions in File Explorer.	2
4	Make a folder with your name and a Word file in it. Rename the file.	4
5	Open the Properties of the file. Write its type, location and size.	4
6	Copy the folder to a pen drive as a backup.	3
7	Write the file system of drive C:.	1
	Total	**20**

★ TEACHING TIPS

- The bank ATM in town and a pocket calculator are the textbook examples of real time operating systems.
- Let each student type one CLI command. They soon see why the correct syntax matters.
- Turn off the projector and ask students to name every WIMP part they remember.
- Use the school timetable for process management: each lesson gets its time and its place in order.
- For memory management, draw a desk (RAM) and a cupboard (secondary memory). The work is done on the desk.
- Plug in a mouse during the lesson and let the class watch the driver message: plug and play.
- Never write a real password on the board. Use 5.17 to teach good password habits.

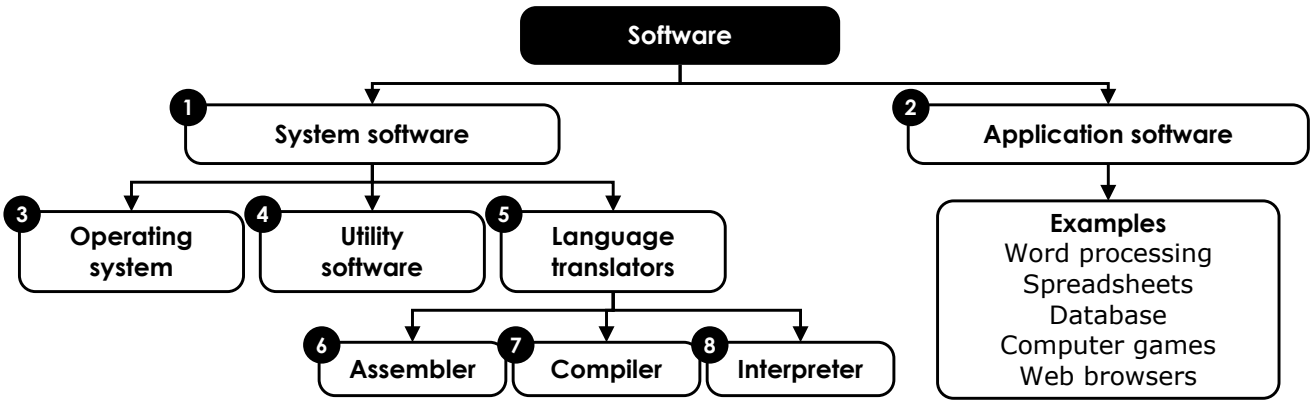
ANSWERS

Chapter 1

The computer system and software

A Complete the software tree

8 marks



1 mark for each box. 6, 7 and 8 may be in any order. Firmware and Hardware are not used.

B Fill in the blanks

8 marks

1. hardware	**2.** firmware	**3.** liveware	**4.** POST
5. Bootstrap	**6.** RAM	**7.** machine	**8.** translators

C Put the booting steps in order

5 marks

a = **4** b = **1** c = **5** d = **2** e = **3**

1 mark each: power on, POST, Bootstrap Loader, OS loaded into RAM, then the OS gets full control (textbook page 139).

D Answer these

4 marks

1. What is firmware? (2 marks)

The booting instructions stored in the ROM (Read Only Memory) (1 mark). The first text shown on the screen when the computer starts is shown by firmware (1 mark).

2. Write two things that the operating system does. (2 marks)

Any two: it manages the hardware and software of the computer; it gives the instructions to install and run application software; it manages the input devices, output devices and memory; it handles the other software. 1 mark each.

TIP: Let students switch on File name extensions on their own computers. Many have never seen them.

E Which part of the computer system?

8 marks

It is ...	Hardware	Firmware	Software	Liveware
The display unit, the keyboard and the mouse	☑	☐	☐	☐
The booting instructions stored in the ROM	☐	☑	☐	☐
Nimal, who operates the library computer	☐	☐	☐	☑
A computer game (a set of instructions)	☐	☐	☑	☐
The hard disk, the speaker and the printer	☑	☐	☐	☐
The first text on the screen at start-up	☐	☑	☐	☐
A word processing program	☐	☐	☑	☐
The users who operate the computer	☐	☐	☐	☑

ANSWERS

Chapter 2

Types, services and user interfaces

A Which type of operating system?

8 marks

It shows ...	Single user	Multi-user	Multi-tasking	Real time
MS DOS serves one person at a time.	☑	☐	☐	☐
An ATM machine gives the output at once.	☐	☐	☐	☑
Many users share one mainframe computer.	☐	☑	☐	☐
Sunil types while music plays on Windows 8.	☐	☐	☑	☐
A calculator shows the answer at once.	☐	☐	☐	☑
Many users use Windows Server together.	☐	☑	☐	☐
Mac OS runs many programs at the same time.	☐	☐	☑	☐
Only one person at a time can use the OS.	☑	☐	☐	☐

B Label the screens

8 marks

1. Command prompt	**2.** Command	**3.** Output	**4.** Cursor
5. Windows (a window)	**6.** Icons	**7.** Menus (a menu)	**8.** Pointer

1 mark each. Accept "prompt" for 1. Parts 5 - 8 are the four WIMP components (Figure 5.4).

C Match them

6 marks

1. Single user OS → C	**2.** Multi-user OS → F	**3.** Multi-tasking OS → A
4. Real time OS → D	**5.** CLI → E	**6.** GUI → B

D Choose the correct answer

4 marks

1. (d) typing documents	**2.** (b) command prompt
3. (b) Windows Server	**4.** (a) Windows, Icons, Menus, Pointer

E True or false

6 marks

1. **T - The operating system manages the hardware and the other software of a computer.**
2. **T - Executing application software is one of the three main services of an OS.**
3. **T - The command line interface was used by early operating systems.**
4. **F - In WIMP, the letter M stands for Mouse.**
5. **F - A multi-tasking operating system can run only one task at a time.**
6. **F - MS DOS is a multi-user operating system.**

ANSWERS

Chapter 3

Managing the hardware and files

A Which management task?

8 marks

1. Process management	**2.** Memory management	**3.** Device management	**4.** Security management
5. File management	**6.** Network management	**7.** Process management	**8.** Security management

1 mark each (the six management tasks of 5.10).

B Read the Properties window

6 marks

1. What type of file is it? **A Microsoft Word document (.docx)**
2. Which program opens it? **Microsoft Word**
3. Where is it saved (its location)? **D:\Grade 10\Science**
4. What is the size of the file? **18.5 KB (18,944 bytes)**
5. How much memory did the storage allocate to save it? **20.0 KB (20,480 bytes)**
6. When was the file created? **March 9, 2026, 8:15:20 AM**

1 mark each. The size on disk is bigger than the size: the storage allocates whole blocks.

TEACHING TIP

- Explain sequential access with a song list on a cassette tape: you must play it in order.

C Sequential or random access?

6 marks

Which access method?	Sequential	Random
Magnetic tape	☑	☐
A hard disk	☐	☑
RAM	☐	☑
A DVD	☐	☑
Reading records 1, 2, 3 ... in order to reach record 50	☑	☐
Going straight to record 50	☐	☑

D Complete the file systems table

5 marks

File system	Maximum capacity of a file	File security	Operating systems
FAT16	**2 GB**	No	MS-DOS, Windows 95 / 98, Linux
FAT32	**4 GB**	**No**	Windows ME / 98, Linux, Mac OS
NTFS	**16 TB**	**Yes**	Windows XP / 7 / 8 / 10, Mac OS, Linux (the default for Windows)

1 mark for each bold cell (textbook page 148).

E Read the file names

10 marks

File	Extension	File type
Nimal.docx	**docx**	**Word**
marks.xlsx	**xlsx**	**Excel**
setup.exe	**exe**	**Executable**
Science Day.pptx	**pptx**	**PowerPoint**
library.accdb	**accdb**	**Access**

1 mark for each extension and 1 mark for each file type.

★ TEACHING TIPS

- Give pairs a printed Properties window. Ask for the location, the size and the size on disk.
- Explain random access with the index of a book: you turn straight to the page you need.
- Open the Properties of drive C: and of a pen drive to show NTFS and FAT32 side by side.
- Say it every lesson: partition first, then format, then install the OS, then save the data.
- Never let students format or partition a drive in class. Show the windows on the projector only.
- A story printed in parts on different pages of a newspaper is a good picture of fragmentation.

ANSWERS

Chapter 4

Utility software, drives and disks

A Read the drives window

8 marks

1. How many hard disk drives does this computer have? **3**
2. Which drive has the most free space? **Local Disk (C:) - 18.8 GB free**
3. What is the total size of drive D:? **117 GB**
4. What is the letter of the pen drive (Removable Disk)? **K:**
5. How much space is used on the pen drive? **2.40 GB (7.18 - 4.78)**
6. Which drives are devices with removable storage? **F:, G: and K:**
7. Partitions **C:** and **D:**; the DVD drive **E:** (2 marks)

1 mark each for 1 - 6. Question 7 follows the textbook rule: the partitions are named C:, D: ... in order, then the CD / DVD drives get the next letters.

B Where is the file?

4 marks

1. Write the location of OS notes.docx. **D:\Grade 10\ICT**
2. Write the location of plants.pptx. **D:\Grade 10\Science**
3. Write the location of marks.xlsx. **C:\Users\Kasuni**
4. Which file is a PowerPoint file? **plants.pptx**

C Which utility?

8 marks

1. Backup software	**2.** Screen saver	**3.** Task manager	**4.** File / data compression
5. Clipboard	**6.** Disk scanner	**7.** Anti-virus software	**8.** Disk defragmentation

D Name the disks

5 marks

1. after formatting	**2.** fragmented	**3.** before partitioning	**4.** defragmented	**5.** after partitioning

TEACHING TIPS

- Show the Defragment and Optimize Drives window, but click only Analyze.
- Ask students to count the drives on the lab computers and to explain their letters.
- Revise with the crossword in pairs: one reads the clue, the other writes the word.

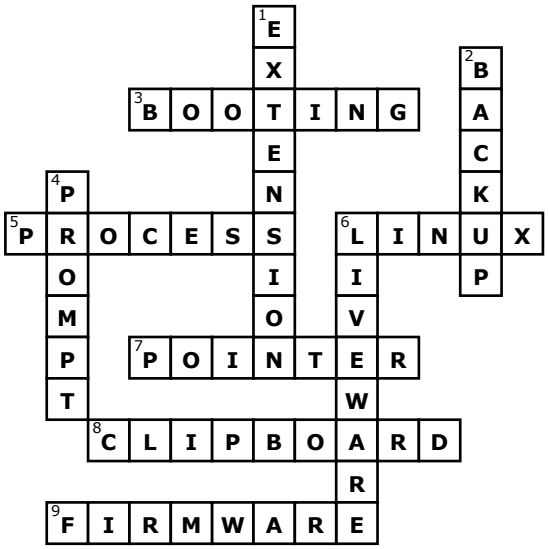
E True or false

8 marks

1. **T - In the past, utility programs had to be bought and installed separately.**
2. **F - If a hard drive is not partitioned, it is labelled as the D: drive.**
3. **T - In Linux, the partitions and flash drives are shown as directories (folders).**
4. **T - When a hard disk is partitioned again, all the data on it is deleted.**
5. **F - USB flash drives must always be formatted before they are used for the first time.**
6. **F - Formatting a drive keeps all the data in it.**
7. **T - A fragmented file takes longer to read.**
8. **F - Linux based operating systems need a defragmentation utility.**

F Crossword

10 marks



Across: 3 BOOTING, 5 PROCESS, 6 LINUX, 7 POINTER, 8 CLIPBOARD, 9 FIRMWARE

Down: 1 EXTENSION, 2 BACKUP, 4 PROMPT, 6 LIVEWARE

TIP: Let pairs swap books and check the tick tables before you mark them.

MARKING SCHEME

Unit 5 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (c) liveware	**2.** (c) Ubuntu
3. (c) a web browser	**4.** (a) Windows Server
5. (b) Icons	**6.** (b) Excel
7. (a) magnetic tape	**8.** (c) A and C only

2 Fill in the blanks

8 marks (1 each)

1. environment	**2.** RAM	**3.** translators	**4.** real
5. syntax	**6.** play	**7.** clipboard	**8.** saver

3 Name the pictures

8 marks (1 each)

1. CLI	**2.** GUI	**3.** Partitioned hard disk	**4.** Fragmented hard disk
5. Sequential access	**6.** Random access	**7.** File access permissions	**8.** Multi-user OS

Firewall and Device driver are not used. Accept "command line interface" for 1 and "graphical user interface" for 2.

4 Answer these

16 marks

1. Write the four steps of the booting process in order. (4 marks)

(1) Electrical power is supplied to the processor, RAM and ROM. (2) The BIOS in ROM performs the POST. (3) The BIOS loads and executes the Bootstrap Loader. (4) The Bootstrap Loader loads the OS into RAM and the OS gets full control. 1 mark for each step in the right place.

2. Name the three main services of an operating system. (3 marks)

Providing a user interface; managing the hardware of a computer; executing application software. 1 mark each.

3. Look at the file name Kamal.pptx. Write (a) its extension, (b) its file type and (c) what the OS uses to identify the type of a file. (3 marks)

(a) pptx (b) a PowerPoint file (c) the file extension. 1 mark each.

4. Write two differences between the FAT32 and NTFS file systems. (2 marks)

Any two: maximum capacity of a file 4 GB / 16 TB; no file security / file security; NTFS is the default for Windows. 1 mark each.

5. Why should you make a backup before you partition or format a disk? (2 marks)

Partitioning again or formatting deletes all the data in the drive (1 mark), so the backup copy keeps the data safe (1 mark).

6. What is a process? Write one thing that the OS allocates to each process. (2 marks)

A running computer program, or a part of it (1 mark). Any one: CPU time, memory, input and output devices (1 mark).

WHAT STUDENTS USUALLY GET WRONG

- They mix up firmware and software. Firmware = the booting instructions stored in the **ROM**.
- They put the booting steps in the wrong order: power on, POST, Bootstrap Loader, OS loaded into RAM.
- They call utility software application software. Utilities are **system software**.
- They mix up multi-user (many users at the same time) and multi-tasking (many tasks at the same time).
- In WIMP they write Mouse for M. M = **Menus** and P = **Pointer**.
- They mix up size and size on disk. Size on disk is the memory allocated by the storage.
- They format before they partition. First **partition**, then **format**.
- They write "defragmented" for "fragmented" (the textbook does it too on page 154).
- They say an application runs from the hard disk. The OS first loads it into **RAM**.

HOMEWORK AND EXTRA PRACTICE

- Textbook Activity questions 2, 3 and 14 (pages 156 - 157): file systems, access methods, FAT32 and NTFS.
- Make a poster of the booting process with the words BIOS, ROM, POST, Bootstrap Loader and RAM.
- Draw the folders of the home or school computer as a tree. Write the location of three files.
- Question 2 also asks about ext4 and ReiserFS, which the chapter does not explain: both are file systems used by Linux.
- Question 11 says "partitioning", but page 155 explains why Linux needs no **defragmentation**. Ask it that way (workbook 5.23).

TEACHING TIPS

- Past O/L papers ask about the booting steps and utility programs. Give one short question each lesson.
- Keep a poster of the six management tasks on the wall during the whole unit.
- Students who finish early can write two more questions for the oral quiz.
- Mark the unit test together the next day. Students learn most from the questions they got wrong.
- Praise answers that use the textbook words: firmware, POST, Bootstrap Loader, process, partition.
- Ask the class to explain the drive letters of a lab computer with a pen drive connected.